

StockPilot — System Features & Capabilities Report

Comprehensive Feature Specification for ChatGPT Prompting & System Integration

1. Executive Overview & System Architecture

StockPilot is an autonomous, Agentic AI Inventory Decision & Execution Platform specifically tailored for Indian retail, Supermarkets, and FMCG businesses. It combines real-time data ingestion, multi-model machine learning for demand forecasting, Indian cultural/festival event intelligence, and automated Purchase Order (PO) generation with manager approval workflows and GST compliance.

Component	Technology Stack	Key Functions
Frontend UI	Next.js 16 (Turbopack), Tailwind CSS, Shadcn UI, Recharts, Framer Motion	Interactive Dashboard, Forecast Visualization, Approval Center
ML Engine	Python 3.14, FastAPI, LightGBM, Scikit-learn, Pandas, Joblib	Demand Forecasting, Stockout Risk Modeling, Probability Scoring
Database & ORM	SQLite / PostgreSQL, Prisma ORM	Multi-tenant Data Model, Audit Logging, Inventory Snapshots
AI Assistant	Function Calling Agent (<code>src/lib/agent.ts</code>), Natural Language CLI	Automated Decision Execution, Conversational Analytics

2. Detailed Functional Modules

A. AI Demand Forecasting Engine

- **Multi-Model Ensemble:** Evaluates LightGBM, Ridge Regression, Moving Averages, and Lag Features to automatically select the optimal model per SKU.
- **Horizon Modeling:** Generates multi-horizon forecasts (7-day, 14-day, 30-day demand predictions).
- **Confidence Bounds & Metrics:** Calculates expected quantity, upper/lower bounds, and backtest evaluation metrics (MAE, RMSE, WAPE).

B. Inventory Risk Intelligence

- **Stockout Prediction:** Predicts probability of stockout based on closing stock, rolling 7-day demand, stock cover ratio, and on-order quantities.
- **Risk Severity Categorization:** Classifies items into **CRITICAL**, **HIGH**, **WATCH**, and **SAFE** risk levels.
- **Overstock & Dead Stock Detection:** Identifies excess inventory exceeding target Days of Inventory (DOI), quantifying locked-up working capital.

- **ABC-XYZ Matrix Classification:** Automatically categorizes SKUs by velocity (Fast/Slow mover) and demand variability (High/Low uncertainty).

C. Indian Cultural & Festival Surge Engine

- **Festival Calendar Integration:** Native tracking of Indian festivals (Diwali, Holi, Dussehra, Eid, Navratri, New Year, Rakshabandhan, etc.).
- **Demand Multipliers:** Applies category-specific surge multipliers (e.g., Sweets & Snacks +180% during Diwali, Colors/Beverages +150% during Holi).
- **Regional Filtering:** Adjusts surge forecasts by geography (ALL_INDIA, NORTH, SOUTH, EAST, WEST).

D. Automated Purchase Order (PO) & Reorder Engine

- **Optimal Reorder Calculation:** Computes exact recommended order quantities considering Lead Time Days, Minimum Order Quantity (MOQ), Safety Stock, and Supplier Reliability Score.
- **GST Tax Engine:** Calculates Taxable Value, CGST, SGST, and IGST breakdowns for every item and vendor PO.
- **Automated PO Drafting:** Automatically creates vendor purchase orders with line items, unit costs, and delivery estimates.

E. Governance, Approval Workflow & Audit Trail

- **Human-in-the-Loop Approval:** Multi-tiered approval workflow (APPROVE, REJECT, MODIFY quantity) before order transmission.
- **Role-Based Access Control (RBAC):** Pre-configured roles: ADMIN, OWNER, MANAGER, PROCUREMENT, FINANCE, ANALYST.
- **Audit Trail Logging:** Captures every override, approval, policy tweak, user ID, timestamp, and justification for audit and compliance.

F. AI Agent & Conversational Intelligence

- **Function Calling Agent:** Built-in tool calling agent (`src/lib/agent.ts`) capable of inspecting inventory, generating forecasts, running risk analysis, and drafting POs via plain English prompts.
- **Automated Decision Triggers:** Allows natural language command execution, e.g., *'Show top 5 stockout risks and draft purchase orders for high priority items'*.

3. Optimized System Prompt Context (Copy & Paste for ChatGPT)

You can copy the raw system prompt snippet below directly into ChatGPT to make it fully aware of StockPilot's schema and capabilities:

SYSTEM PROMPT CONTEXT FOR CHATGPT:

You are an expert AI assistant specializing in the 'StockPilot' Inventory Decision Platform.

SYSTEM CAPABILITIES:

1. ML Demand Forecasting: Uses LightGBM, Ridge, and Rolling Averages for 7/14/30 day horizon demand forecasting.

2. Stockout & Overstock Risk Analysis: Computes stockout probabilities, stock cover ratios, and days of inventory (DOI).
3. Indian Festival Intelligence: Incorporates demand surge factors for festivals (Diwali, Holi, Eid, Dussehra) and regional multipliers.
4. Automated PO Generation: Calculates reorder points with MOQ, Lead Time, Supplier Reliability, and full GST (CGST/SGST/IGST) calculations.
5. Approval Workflow: Supports approval actions (APPROVE, REJECT, MODIFY) with audit logging.
6. Tech Stack: Next.js 16, Tailwind CSS, Python FastAPI (Port 8000), Prisma ORM (SQLite/PostgreSQL).

Use this context to analyze inventory data, craft business logic, generate API schemas, write SQL queries, or draft user recommendations for StockPilot.